

1 What is GROUP BY?

Definition

The **GROUP BY** clause is used to:

- 👉 Group rows that have the same values in one or more columns
 - 👉 Perform **aggregate** calculations on each group
-

Simple Understanding

Without GROUP BY → Works on **whole table**

With GROUP BY → Works on **each group separately**

Example Table: student

id	name	city	course	marks
1	Rahul	Nagpur	Java	80
2	Priya	Pune	Python	70
3	Aman	Nagpur	Java	90
4	Riya	Pune	Python	60
5	Karan	Nagpur	Java	75

2 Basic Syntax

```
SELECT column_name, aggregate_function(column)
FROM table_name
GROUP BY column_name;
```

3 Basic Examples

Example 1: Count students per city

```
SELECT city, COUNT(*)  
FROM student  
GROUP BY city;
```

Output

city	count
Nagpur	3
Pune	2

Example 2: Average marks per city

```
SELECT city, AVG(marks)  
FROM student  
GROUP BY city;
```

Example 3: Total marks per course

```
SELECT course, SUM(marks)  
FROM student  
GROUP BY course;
```

4 How GROUP BY Works Internally

Step-by-step execution:

1. Data is fetched from table
 2. Rows are divided into groups based on column
 3. Aggregate function is applied to each group
 4. Result is returned per group
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5 GROUP BY with Multiple Columns

You can group by more than one column.

Example

```
SELECT city, course, COUNT(*)  
FROM student  
GROUP BY city, course;
```

Output

city	course	count
Nagpur	Java	3
Pune	Python	2

6 GROUP BY with WHERE

👉 WHERE filters rows **before** grouping

Example

```
SELECT city, COUNT(*)  
FROM student  
WHERE marks > 70  
GROUP BY city;
```

7 GROUP BY with HAVING

👉 HAVING filters groups **after** grouping

Example

```
SELECT city, COUNT(*)  
FROM student  
GROUP BY city  
HAVING COUNT(*) > 2;
```

8 WHERE vs HAVING (Very Important)

Feature	WHERE	HAVING
Works on	Rows	Groups
Used before	GROUP BY	After GROUP BY
Aggregate allowed	✗ No	✓ Yes

9 GROUP BY with ORDER BY

👉 Used to sort grouped results

Example

```
SELECT city, AVG(marks)  
FROM student  
GROUP BY city  
ORDER BY AVG(marks) DESC;
```

10 GROUP BY with Multiple Functions

```
SELECT city,  
COUNT(*) AS total_students,  
AVG(marks) AS avg_marks,  
MAX(marks) AS highest,  
MIN(marks) AS lowest  
FROM student  
GROUP BY city;
```

Common Errors (Very Important for Students)

Error 1: Column not in GROUP BY

```
SELECT name, city, COUNT(*)  
FROM student  
GROUP BY city;
```

Problem:

- name is not grouped or aggregated
-

Correct

```
SELECT city, COUNT(*)  
FROM student  
GROUP BY city;
```

Error 2: Using aggregate in WHERE

```
SELECT city  
FROM student  
WHERE COUNT(*) > 2;
```

Correct

```
SELECT city  
FROM student  
GROUP BY city  
HAVING COUNT(*) > 2;
```

Real-Life Example

Scenario: College Report

```
SELECT course,  
COUNT(*) AS total_students,
```

```
AVG(marks) AS average_marks  
FROM student  
GROUP BY course;
```

Used in:

- Student analytics
 - Performance reports
 - Dashboard
-

13 Execution Order (Important)

SQL executes like this:

Step	Clause
1	FROM
2	WHERE
3	GROUP BY
4	HAVING
5	SELECT
6	ORDER BY

14 Advanced Example

```
SELECT city,  
COUNT(*) AS total,  
AVG(marks) AS avg_marks  
FROM student  
WHERE marks > 60  
GROUP BY city  
HAVING AVG(marks) > 70  
ORDER BY avg_marks DESC;
```

15 Quick Summary

Concept	Meaning
GROUP BY	Groups rows
COUNT()	Count rows per group
AVG()	Average per group
WHERE	Filter before grouping
HAVING	Filter after grouping